



Jennifer C. Cremer

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Education

University of Florida, Herbert Wertheim College of Engineering

Gainesville, FL

Ph.D, Computer Graphics & Visualization

2018 – 2026

Relevant Courses: Advanced Computer Graphics, Multimodal Data Mining

MSc, Computer Science

2018 – 2021

Relevant Courses: Concurrent Programming, Computer Graphics, Computational Geometric Modeling, Information Visualization

BS, Digital Arts & Science

2014 – 2018

External focus in Applied Physics: Statics, Thermodynamics, Fluid Mechanics

Relevant Courses: Human-Computer Interaction, Fundamentals of Computer Graphics, Design Patterns, Obj-Oriented Programming, Theory & Practice of Multimedia Production, 3D Modeling & Animation, Operating Systems

Research

Graduate Student Researcher

2018 – 2026

Scan2Twin – Jörg Peters: *SurfLab* & **Eric Ragan:** *INDIE Lab, Dept. of CISE, UF*

- Refactored orphaned VR software to adhere to proper data handling, object-oriented paradigms, and modern UX guidelines
- Developed C++/OpenGL software for virtual reality (VR) to voxelate medical images and trace out vessels as B-spline curves
- Converted project base from C++/OpenGL/OpenVR to the Unity3D Engine and C#
- Developed VR toolsets for spatial understanding and interactive modeling of voxel data
- Demonstrating functionality and visualization prototypes of anatomy to surgical teams
- Managing project definitions and scope with collaborators in the Colorectal Oncology Team at UF Health: Shands
- Integrated Segment Anything using PyTorch into Unity to assist in volumetric noise reduction
- Integrated responsive shrink-wrapping methods to transform voxel clouds into discrete meshes
- User Study on the impacts of modeling experience on efficiency and accuracy of mesh editing in VR and desktop interfaces
- Investigated capabilities of Skeleton Extraction methods using Pytorch to improve meshing process
- Mixed-Factorial User Study exploring the aptitude of various Volumetric Rendering techniques given different investigative tasks

Redirected Walking Study – Brett Benda: *INDIE Lab, Dept. of CISE, UF*

2024

- Assisted execution and analysis of a Between-Subject VR Study investigating impacts of technique awareness on translation gain detection and thresholds

Undergraduate Researcher

2017 - 2018

TIPS – Jörg Peters: *SurfLab*

- Developed a custom file format to store mesh data as well as scene hierarchy relationships
- Created a Python script for Blender to parse custom files of mesh data and object hierarchy into soft-body surfaces with spring constraints
- Developed user-report interfaces for surgical simulation software with screen captures and descriptions of key training moments using Qt
- Acted as interlocutor for an interdisciplinary team with the UF Veterinary School

Conferences & Papers

Virtual Reality Enabled Segmentation for Volumetric Analysis of Responses to Neoadjuvant Therapy for Rectal Cancer

American Society of Colon and Rectal Surgeons 2026 (Extended Abstract)

Christine Rodhouse MD, Grace Thompson MD, **Jennifer Cremer**, Jorg Peters PhD, Krista P Terracina MD.

Immersive VR 3D Model for Rectal Cancer Surgical Planning

Diseases of the Colon and Rectum 2026 (Journal)

Pavel Mazirka MD, **Jennifer Cremer**, Jeremy Balch MD, Johan Nordenstam MD PhD, Thomas E Read MD, Joseph R Grajo MD, Jorg Peters PhD, Krista P Terracina MD.

Empirical Study of Virtual Reality and Desktop Systems for Qualitative Editing of 3D Meshes: Impacts of Expertise and Context

ACM Symposium on Virtual Reality Software and Technology 2025 (Conference)

Jennifer C. Cremer, Connor Lausch, Jorg Peters and Eric Ragan

Improving Radiology Communications and Patient Trust with Virtual Reality

ACM Symposium on Virtual Reality Software and Technology 2025 (Poster Extended Abstract)

Cremer, J., Lausch, C., Peters, J., Terracina, K., and Ragan, E.

Scan2Twin: Virtual Reality for Enhanced Anatomical Investigation

IEEE Conference on Virtual Reality and 3D User Interfaces (IEEEVR 2024) (Doctoral Consortium)

Jennifer C. Cremer

Immersive VR 3D Model for Rectal Cancer Robotic Surgery

American Society of Colon and Rectal Surgeons via Intuitive Research - 2023 - Video Abstract

P. Mazirka, **J. Cremer**, J. Balch, A. Rashid, K. Ehresmann, L. Goldstein, J. Nordenstam, T.E. Read, J. Grajo, J. Peters, K. Terracina

Patient-Specific MRI VR Model Construction and Simulation

Women in Scientific Computing on Complex Physical and Biological Systems (Poster) - 2022 - Gainesville, FL

Jennifer C. Cremer, Jörg Peters

From Scans & Model Collections to Interactive Surgical Simulation

ACS Surgeons and Engineers 2021 - Poster

Jennifer Cremer, Ruiliang Gao, Krista Terracina MD, Jörg Peters

VascularVR (Research Exhibitor)

Academic Surgical Congress 2020 - Orlando, FL

Grants & Awards

Research in Robotic Technology Grant - Research Foundation of the ASCRS	2021-2023
CISE Department Nominee, Outstanding Graduate Teaching Assistant Award, UF	2020
NSF GRFP Honorable Mention, Computer Graphics and Visualization	2020

Volunteering & Professional Service

Academy Software Foundation: Diversity & Inclusion Working Group	2020 - Present
SIGGRAPH Birds of a Feather	2025
SIGGRAPH Student Volunteer Info Session	2024 – 2026
Summer Learning Program Organization Team	2023 – 2025
DevDays Volunteer	2023
Student Volunteer for ACM SIGGRAPH	2023, 2024
VFX Careers Webinar Series, Academy Software Foundation	2021
University panelist for “VFX Careers: Technical Director”	
Lead presenter for “University Content: Building from Source with Cmake”	
Understudy to Japan Study Abroad Organization	2019 - 2022
Teaching Assistant consultant for CISE/Cross-Cultural Engineering Abroad Kyoto, Japan	
Student Consultant to Digital Arts and Science Program	2018 - 2019
Organization and course direction suggestions to present DAS Director (Joshua Fox)	

Teaching

Dept. of Computer & Information Science & Engineering, Univ. of FI

Teaching Assistant

2017 - 2025

- Assisted the professor with course management and individual student affairs
- Mentored students through office hours and one-on-one communication
- Developed project specifications, lab assignments, and grading protocols
- Taught lab sections covering review and supplemental materials

Courses:

COP4533 – Algorithm Abstraction and Design	2025
COP3530 – Data Structures and Algorithms	2025
COP3503 – Programming Fundamentals 2 in C++	2019, 2020, 2024, 2025
CAP3027 – Introduction to Computational Media	2024
CAP4053 – Artificial Intelligence for Computer Games**	2019, 2024
CAP5705 – Computer Graphics**	2022
CIS6930 – Information Visualization	2022
COP3504C – Adv. Programming Fundamentals**	2021
CIS4930 – Special Topics: Performant Programming in Python**	2020, 2022
COP4600 – Operating Systems**	2017–2020

***Developed original assignment specifications and materials*

UF at Kyoto University Summer Abroad Program

2022

CIS4930 – Performant Programming in Python | CIS4930 – Cross-Cultural Engineering

Instructor on record

2020 - 2021

CIS4930 – Performant Programming in Python

2021

- Instructed students on industry soft-skills including team dynamics and communication of ideas
- Saw to the needs of both traditional student and online student sections
- Coached course assistants in observing students for troubling behavior and mediation technique

CIS4930: Special Topics in CISE - Design Patterns in OOP

2020, 2021

- Original development of course content and material on key paradigms in programming and efficient design choices
- Instructed students on industry soft-skills including team dynamics and communication of ideas to an interdisciplinary audience
- Executed a smooth transition to online platform at the start of the covid-19 pandemic

Mentoring

SURFLab Undergraduate Student Research Coordinator

2018 - 2026

Dept. of Computer & Information Science & Engineering, Univ. of FI

Project manager to multiple mixed groups of students working on both independent projects and dissertation subprojects.

Provided guidance and oversight on reading research papers, project organization, presenting research ideas to the group, and developing mature scientific practices.

Team Lead, Academy Software Foundation Summer Learning Program

2021, 2022

Worked with a small group of learners over the course of the program to stay on track, grow as a cohort, and optimize their meetings with their industry mentors.

2021: Ximena Jaramillo, JaNiece Campbell, Jessica Zhou, Linda Lam

2022: Parag Gupta, Stephanie Lim

Operating Systems Team Lead

2018 - 2020

Dept. of Computer & Information Science & Engineering, Univ. of FI

- Trained the staff on the responsibilities and general approach to being both a TA and mentor
- Provided detailed instruction on how to conduct discussion sections
- Led team of peer mentors and teaching assistants in organization of discussion content
- Taught the staff innovative strategies to better educate others, including utilizing the Socratic Method in Office Hours
- Coached graduate students to serve as mentors to the students under their care from a holistic standpoint in terms of both student major as well as each student's unique perspective

Summer Science Training Program Mentor, UF

2017, 2019

- Introduced high school students to the environment and experience of working in a university research lab.
- Outlined project milestones and developed a comprehensive timeline for tasks.
- Provided guidance in conducting background research and how to frame accomplishments into an organized presentation.

Technical Skills

Programming Languages:

C++, C#, Python, Java, OpenGL, WebGL, HLSL, Bash, JavaScript

Software Packages & Tools:

Unity 3D, OpenXR, Qt, PyTorch, Adobe CC Suite, Autodesk Maya, Blender

Soft Skills

Mentorship, Interdisciplinary Communication, Project Management, Team Management, Feature Definition

Extracurricular

WarHammer Miniature Painting

2023 – Present

Advanced Open-Water SCUBA Diving – PADI certification

2016 – Present

Amateur Wildlife Photography – birds in flight & macro bugs

2012 – Present